Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS at USC with 2+ years building and supporting **data-driven operational software**, **SQL-backed applications**, and automation-heavy workflows, with experience in debugging, validation, and AI-enabled process improvement.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Designed and optimized SQL-backed application workflows supporting 500+ users, reducing latency from **1.2s to 350ms** through query tuning, indexed access patterns, and structured backend improvements.
- Built ETL-style integration pipelines processing 5,000+ monthly jobs with validation, reconciliation, and CI/CD-backed releases, improving operational reliability and accelerating issue resolution across production systems.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed and tested full-stack application features handling 10,000+ daily requests, translating product needs into maintainable backend services and interfaces while improving debugging, release confidence, and documentation quality.
- Implemented secure batch-processing workflows for PDF, DOCX, and ZIP ingestion with AWS-backed persistence, reducing feedback turnaround by **60%** while supporting stable production operations and issue triage.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built Python-based experimentation pipelines across 100,000+ records, creating reproducible evaluations and technical artifacts that improved debugging quality, analysis consistency, and communication of system behavior.
- Produced technical reports and targeted validation scenarios translating findings into actionable improvements, supporting stronger design reviews, clearer handoffs, and better engineering decisions across collaborators.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered API and ETL microservices using Python, AWS Glue, and S3, improving response time by **30%** while supporting scalable data movement, structured delivery, and reliable production workflows.
- Maintained 20+ automated test suites using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and improving confidence during collaborative release cycles.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built an operations platform with WebSockets and REST APIs, supporting 2M+ synchronized records and six workflows through modular services, responsive interfaces, and maintainable system integrations.
- Deployed Azure SQL with Jenkins-based CI/CD and automated testing, achieving 95%+ successful build validation while improving release consistency and production-readiness across delivery cycles.

Skills

Languages: **C#** familiarity, **SQL**, Python, Java, JavaScript/TypeScript, C++
Software & Data: WinForms/WPF familiarity, Visual Studio, SQL Server, PostgreSQL, MySQL, data validation, root cause analysis, debugging
Automation & AI: CI/CD, Jenkins, automated test suites, ETL pipelines, AI-assisted development, process automation, data analytics
Platforms: Git/GitHub, Azure SQL, AWS, Docker, Agile/Scrum, operational support, cross-functional collaboration

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases, accelerating schema exploration, query authoring, and structured debugging of large data workflows in enterprise-style environments.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed a containerized RESTful data processing API with automated CI/CD releases and validation checks, improving deployment reliability, interoperability, and supportability for production operational workflows.

FinNexus Code | *Next.js, TypeScript, Kubernetes, MongoDB*
- Built a secure platform supporting 50,000+ users and 1M+ transactions, combining real-time updates, scalable data handling, and operational monitoring to demonstrate high-volume application patterns.